

UTRE — Experimental Validation Plan

Harmonic Nexus Core (HNC) • Illumination Chip • A.L.F.I.E. • EPAS

1. Purpose & Scope

This plan defines the experimental validation required to demonstrate reproducible resonance indices ($\Psi_{\blacksquare}$, ΔM , $R(t)$) across hardware (HNC devices) and humanware (HRV/EEG cohorts), following the UTRE Sections 1–7. All studies are preregistered, multi-site, and open-data with cryptographic integrity seals.

2. Objectives & Hypotheses

Objective	Primary Endpoint	Hypothesis
Hardware: ZPE/DCE extraction	ΔM beyond controls; energy balance and stability	Illumination Chips exhibit statistically significant ΔM and Γ
Humanware: coherence entrainment	$R(t)$ slope increase vs sham	0.1 Hz entrainment yields higher HRV/EEG coherence
Integration: planetary coupling	Cross-correlation with Kp/Ap/Schumann	Device + human ΔM correlate with planetary drivers

3. Study Design

Multi-site, parallel tracks (hardware & humanware), 12-month pilot. Minimum of three independent laboratories per track, with replication required at ≥ 2 sites. Blinding, sham controls, and standardized pipelines for $\Psi_{\blacksquare}$, ΔM , $R(t)$.

4. Metrics & Definitions

Index	Definition	Units / Notes
$\Psi_{\blacksquare}$ (Origin)	Band-limited spectral power above noise	V^2/Hz (devices), $\text{ms}^2/\mu V^2$ (HRV/EEG)
ΔM (Operation)	Shift in coherence or spectral power vs baseline	Unitless z-score / rate changes
$R(t)$ (Trajectory)	Time-derivative/slope of coherence indices	$\Delta \Gamma/\Delta t$ (per min)
Γ (Coherence)	Normalized variance of dominant frequency stability	0–1, target ≥ 0.945
Integrity Index	SNR $\times$ stability duration	Unitless
Resonance Index	Peak sharpness $\times$ phase-lock ratio	Unitless

5. Hardware Protocol (Illumination Chip / DCE Platforms)

Devices: Illumination Chips, Casimir microcavities, SNSPD photon detectors. Baselines: shielded runs, dummy loads, off-resonance modulation.

Procedure	Instrumentation / Settings	Data Products
Gate-A validation	SNSPDs ($g^2(0)$), calibrated timing jitter; Faraday/cryogenic shielding	$g^2(0)$, jitter histograms
Energy balance	Precision power meters (DC/AC), 4-wire shunts, the Planck constant	Power logs, error bounds
Coherence runs	Resonant modulation, coil B-fields, RF drive sweep	Γ , ΔM , $R(t)$, PSDs, waveforms

6. Humanware Protocol (HRV/EEG Entrainment)

Participants: $n \geq 30$ per arm, randomized (entrainment vs sham). Interventions: 0.1 Hz breathing vs non-resonant sham. Data: RR intervals, EEG (alpha), cortisol assays, journaling.

Procedure	Instrumentation / Settings	Data Products
Baseline & sham	ECG (250–1000 Hz), EEG (19–32 ch), saliva cortisol	$\Psi_{\blacksquare}$ (HRV/EEG), baselines
Entrainment	Paced breathing @0.1 Hz, eyes-closed EEG	ΔM , Γ , phase-lock ratios
Recovery & follow-up	Repeat HRV/EEG + cortisol (24–72h)	$R(t)$ slopes, resilience scores

7. Controls, Randomization, Blinding

Hardware: dummy loads, off-resonance runs, thermal-only profiles. Humanware: sham breathing, eyes-open EEG baselines. Randomization stratified by site; analysts blinded to condition labels.

8. Instrumentation & Calibration

- Power: DC/AC meters (traceable), 4-wire shunts, thermal cameras/RTDs.
- RF/EM: spectrum analyzers, lock-ins, B-dot/Hall probes, EMI current clamps.
- Photonics: SNSPD/APD detectors, time-taggers (<50 ps), dark-rate controls.
- Bio: 3-lead ECG, 19–32ch EEG, validated breath pacers, salivary cortisol kits.
- Calibration: daily functional checks; weekly traceable calibrations; inter-site round robin.

9. Data Management & Integrity

All raw time-series, metadata, and analysis scripts are stored in a shared repository (e.g., Supabase/GitHub). Files are hashed (SHA256). Access: aggregate public; raw under DUA. All analyses are containerized for reproducibility.

10. Statistical Analysis Plan

Analysis	Method	Notes
Primary endpoints	Paired/within-subject tests (Welch t, Wilcoxon); effect sizes (Cohen's d)	Multiple comparisons correction for multiplicity
Time-trajectory	Mixed-effects models on R(t); slope comparisons	Random intercepts by subject/site
Planetary coupling	Cross-correlation; surrogate phase randomization	Predefined lags; null via surrogates
Rare-event risk	Monte Carlo ($\geq 10^6$) with variance reduction	Upper confidence bounds reported

11. Validation & Success Criteria

Track	Criteria
Hardware	ΔM replicated in ≥ 2 labs; energy anomalies beyond error bounds; $\Gamma \geq 0.945$
Humanware	ΔM replicated in ≥ 2 cohorts; HRV/EEG coherence > sham ($p < 0.05$)
Integration	Significant R(t) correlation with planetary drivers across tracks

12. Ethics, Safety & Governance

Human studies: IRB approval, informed consent, anonymization. Device tests: hazard analysis (FMEA), HALT/HASS screening. Governance: UTRE Oversight Board; preregistration and open protocols; no clinical claims; non-coercive participation.

13. Timeline & Milestones (12-Month Pilot)

Phase	Months	Milestones
Setup & calibration	0–2	Site onboarding, protocol training, round-robin calibration
Pilot runs	3–5	First $\Psi/\Delta M/R(t)$ datasets; analysis pipeline lock
Replication	6–9	Cross-site replication; interim report
Synthesis & publication	10–12	Hardware + humanware manuscripts; UTRE synthesis preprint

14. Reviewer Checklists

Domain	Checklist (pass/fail)
Design	Endpoints preregistered; controls defined; blinding; replication plan

Hardware	Gate-A, energy balance, EMI guard, cryptographic hashes
Humanware	Randomization, sham, IRB, anonymization
Stats	Power calc, corrections, surrogates, mixed-effects for R(t)
Data	Open repo, versioned code, containerized analysis